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System Design & Software Architecture

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Enterprise Nexus: Multi Agent Business Automation OS

1. Team Information

Team Name	CSE4204-8B-T04
Section	8B
Project Title	Enterprise Nexus: Multi-Agent AI Business Automation OS
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Course	CSE 4204 — Mobile Computing Lab
Submission	Week 04 — System Design & System Architecture
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GitHub	https://github.com/Nazmussakib247/CSE4204-8B-T04-Enterprise-Nexus-Multi-Agent-AI-Business-Automation-OS-with-Security-Intelligence

2. Project Information

2.1 Project Description

Enterprise Nexus is a Multi-Agent AI Business Automation Operating System designed to unify and intelligently automate core business functions for small and medium enterprises, startups, and growing organizations.

The platform deploys five specialized AI agents — an HR Agent, a Finance Agent, a Support Agent, an Analytics Agent, and an Executive Agent — each responsible for a distinct operational domain. These agents operate through a centralized orchestration layer powered by LangChain and CrewAI, enabling them to collaborate autonomously, delegate sub-tasks across departments, and deliver synthesized intelligence to a unified, role-based web dashboard. A dedicated Security Intelligence module monitors threats and enforces data protection across the entire system, ensuring enterprise-grade safety without requiring dedicated security infrastructure.

The system is architected to replace the fragmented landscape of disconnected SaaS tools that currently force businesses to manage HR, finance, customer support, and analytics through separate platforms with no intelligent cross-domain layer. By consolidating these functions into a single cohesive operating system, Enterprise Nexus eliminates manual data transfer, reduces subscription costs, and accelerates decision-making through real-time, AI-driven insights.

2.2 Objectives

Enterprise Nexus is built around six core objectives:

1. **Automate Department-Level Operations** — Deploy five specialized AI agents (HR, Finance, Support, Analytics, Executive) to autonomously handle routine business workflows that traditionally require dedicated human teams.

2. **Deliver Intelligent Decision Support** — Leverage Google Gemini 1.5 Flash/Pro to generate data-driven insights, candidate rankings, anomaly alerts, KPI narratives, and strategic briefings — replacing manual report generation.

3. **Centralize Business Intelligence** — Unify HR, financial, support, and operational data into a single platform with a real-time dashboard, giving decision-makers a 360° view of business health without switching tools.

4. **Ensure Enterprise-Grade Security** — Implement JWT-based authentication, RBAC (Admin/User roles), bcryptjs hashing, rate limiting, input sanitisation, and security headers to protect sensitive business data at every layer.

5. **Enable Scalable, Cloud-Native Deployment** — Build a production-ready system using Docker, GitHub Actions CI, Vercel (frontend), Render (backend), and Supabase (PostgreSQL) — deployable by any team with zero infrastructure expertise.

6. **Demonstrate Full-Stack AI Engineering** — Produce a well-documented, end-to-end software system with 37 REST API endpoints, 8 database tables, 11 frontend pages, and 5 AI agent pipelines as a complete academic deliverable for CSE 4204.

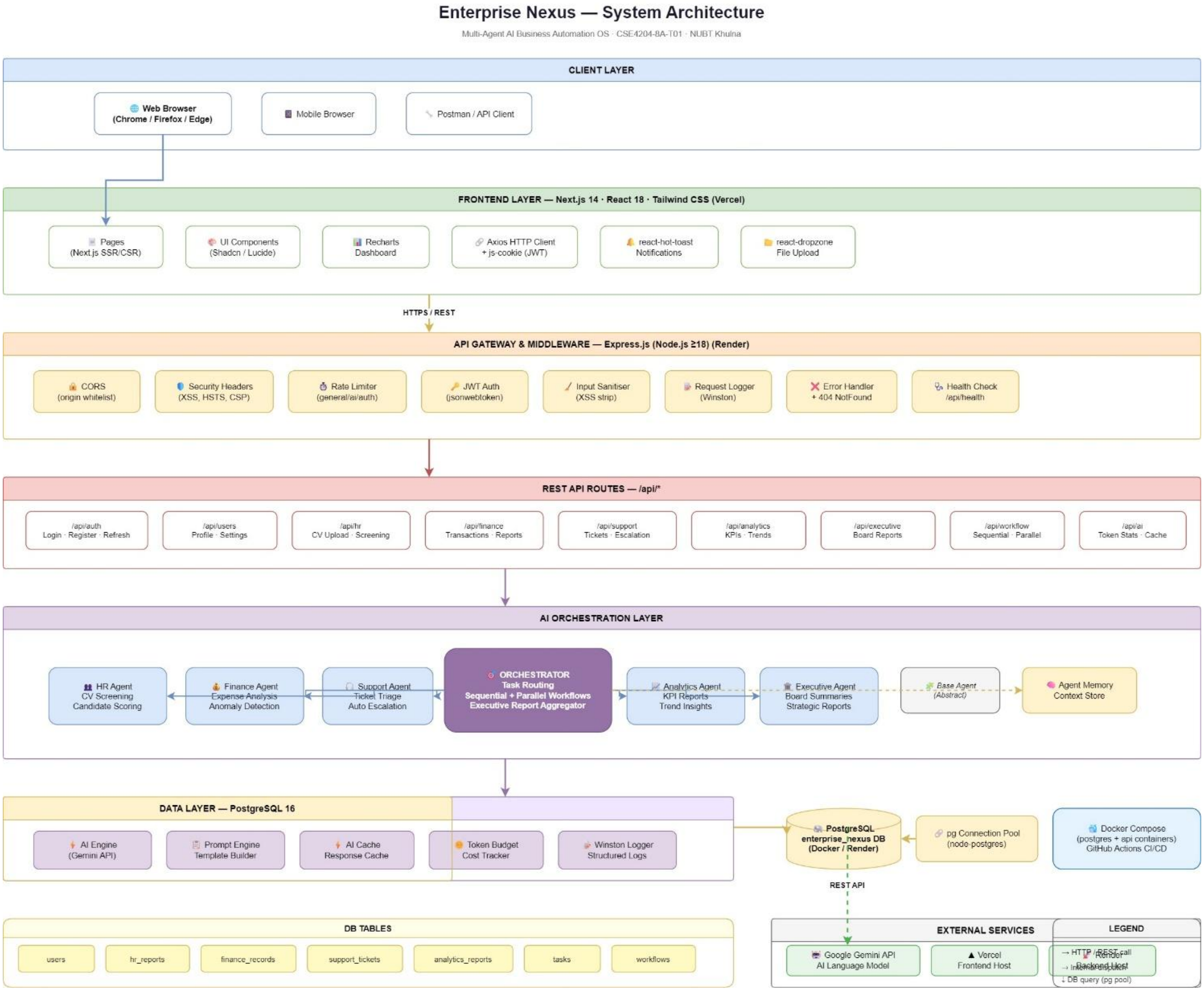
2.3 Proposed Features

- Multi-Agent AI Orchestration (HR, Finance, Support, Analytics, Executive)
- Intelligent CV Screening via the HR Agent
- Financial Pattern Recognition and Anomaly Detection
- Context-Aware Customer Support with RAG
- Executive Synthesis — Cross-domain strategic summaries
- Semantic Search with AI-generated embeddings
- Security Intelligence and Threat Monitoring
- Role-Based Unified Web Dashboard
- Real-Time AI-Driven Insights and Reporting
- Document and Policy Retrieval via Vector Search

3. System Architecture

Enterprise Nexus follows a 7-layer hierarchical architecture that separates concerns from the client browser down to persistent data storage. Each layer communicates with adjacent layers through well-defined interfaces — HTTPS/REST between frontend and backend, WebSocket's for real-time events, n8n webhooks for orchestration triggers, and the Google Gemini AI API for intelligent task processing.

3.1 Architecture Diagram



3.2 Architecture Layer Overview

Layer	Name	Key Technologies	Responsibility
L1	**Client / Browser**	Web Browser	User interaction entry point for all roles
L2	**Frontend**	Next.js 14 App Router, React 18, Tailwind CSS, shadcn/ui	SSR+CSR UI, routing, state, dashboard rendering
L3	**API Gateway**	Express Middleware, JWT Auth, Rate Limiter, CORS	Auth enforcement, security perimeter, token validation
L4	**Backend Services**	Node.js 20 LTS, Express.js 4.x, Socket.io 4.x	Business logic, REST APIs, WebSocket event management
L5	**Orchestration**	n8n (self-hosted), Webhook Triggers, Data Router	Visual workflow automation, agent task routing
L6	**AI Agent**	Google Gemini 1.5 Pro API,	Domain-specific intelligent processing per

Layer	Name	Key Technologies	Responsibility
	Layer	Prompt Engineering	module
L7	**Data Persistence**	PostgreSQL 16, Redis 7, pgvector 0.6, File Store	ACID storage, caching, vector search, file quarantine

3.3 Layer By Layer Description:

L1 — Client / Browser

All users access Enterprise Nexus through a standard web browser. Four human roles interact with the system: Admin (full access), Manager (analytics + finance), HR Officer (talent acquisition), and Support Staff (ticket management). No native mobile app is required — the Next.js frontend is fully responsive.

L2 — Frontend (Next.js 14 App Router)

The frontend is built on Next.js 14 using the App Router paradigm, enabling Server-Side Rendering (SSR) for initial page loads and Client-Side Rendering (CSR) for interactive dashboard components. Each module has a dedicated page: /dashboard, /hr, /finance, /support, /security. Tailwind CSS and shadcn/ui provide the component library. Recharts and D3.js power the analytics visualizations.

L3 — API Gateway & Auth Middleware

Every incoming request passes through the Express middleware stack: (1) CORS validation, (2) rate limiting to prevent abuse, (3) JWT token verification for protected routes. Role-based access control (RBAC) is enforced here — the middleware reads the user's role from the decoded JWT and blocks unauthorized operations before any business logic executes.

L4 — Backend Services (Node.js + Express)

Five independent service modules handle business logic: Auth Service (/api/v1/auth), HR Service (/api/v1/hr), Finance Service (/api/v1/finance), Support Service (/api/v1/support), and Security Service (/api/v1/security). Socket.io manages WebSocket connections for real-time dashboard updates. Each service follows a controller-service-model pattern.

L5 — Workflow Orchestration (n8n)

n8n serves as the intelligent glue between the backend and AI agents. When a user uploads a CV, invoice, or submits a ticket, the backend fires a POST to the n8n webhook. n8n identifies the data type, routes it to the appropriate AI agent workflow, collects the structured result, and returns it to the backend. This decouples AI logic from application code.

L6 — AI Agent Layer (Google Gemini API)

Five specialized AI agents powered by Google Gemini 1.5 Pro: Security Agent (phishing/spam detection), HR Agent (CV screening + semantic scoring), Finance Agent (invoice OCR + anomaly detection), Support Agent (sentiment analysis + RAG response generation), and Executive Agent (cross-domain summary synthesis). Each agent uses carefully engineered prompts specific to its domain.

L7 — Data Persistence Layer

PostgreSQL 16 is the primary relational store for all structured business data. Redis 7 handles session caching and pub/sub messaging — when n8n completes an AI task, it publishes to a Redis channel, and Socket.io consumes this to push real-time updates to dashboard clients. The vector extension on PostgreSQL enables semantic similarity search for RAG-based support responses. A file system store handles CV and invoice file uploads with quarantine zone isolation.

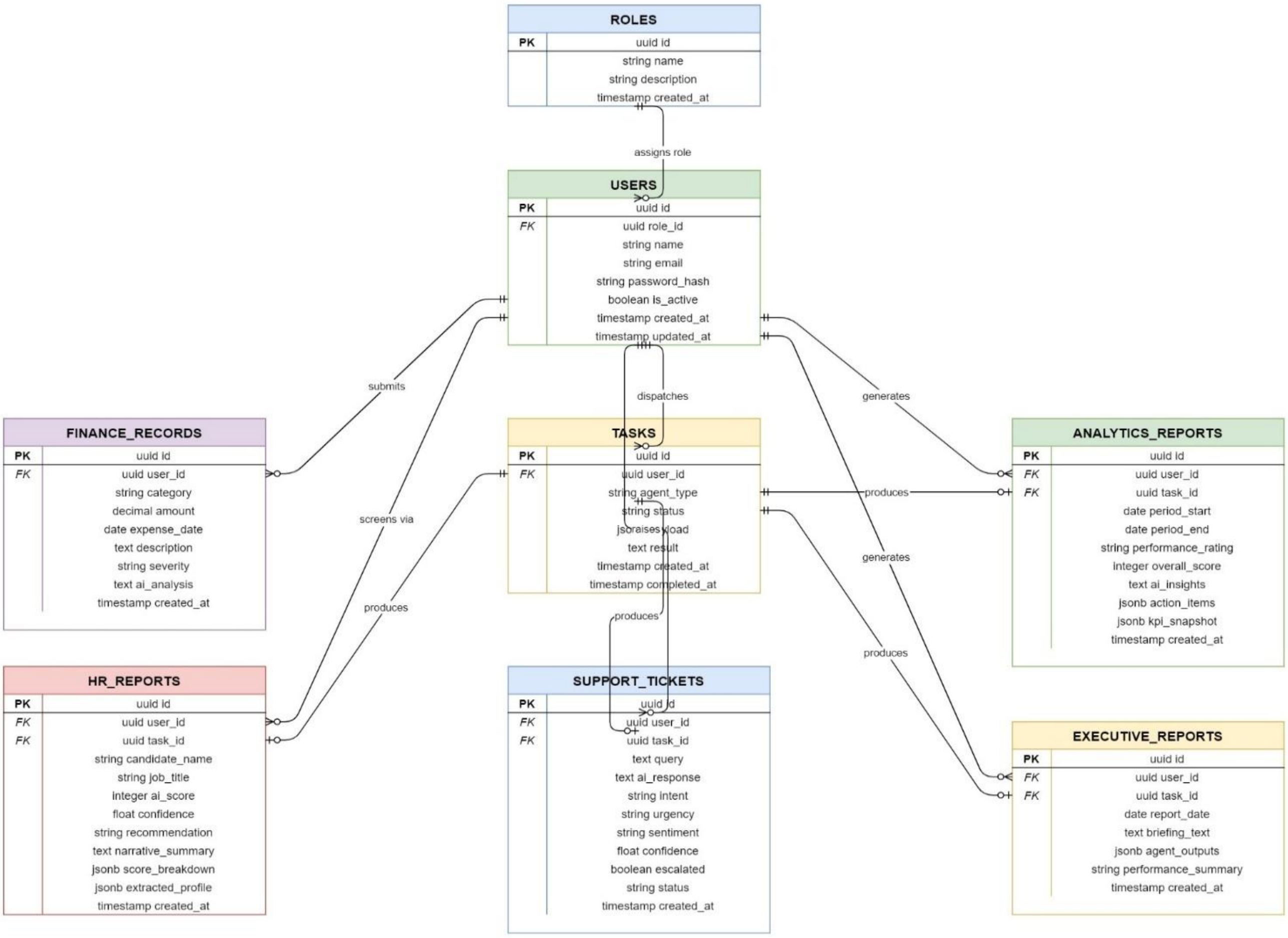
4. Request Flow Architecture

The following sequence illustrates a complete end-to-end request through the system:

Step	From	To	Protocol	Action
1	Browser	Next.js Frontend	HTTPS	User submits document (CV, invoice, ticket)
2	Next.js	Express Backend	REST / POST	Frontend sends multipart form or JSON payload
3	Express	Auth Middleware	Internal	JWT validated, role checked, request authorized
4	Express	PostgreSQL	SQL INSERT	Record created, file saved to storage
5	Express	n8n Webhook	HTTP POST	Task payload dispatched to orchestration layer
6	n8n Router	AI Agent	Gemini API	Appropriate agent receives structured prompt
7	Gemini API	n8n	JSON Response	AI output returned (score, summary, analysis)
8	n8n	Express	HTTP Response	Result forwarded back to backend
9	Express	PostgreSQL	SQL UPDATE	AI result stored, record status updated
10	Express	Redis	PUB/SUB	Event published to real-time channel
11	Redis	Socket.io	Subscribe	Socket.io receives channel message
12	Socket.io	Browser	WebSocket	Dashboard updated in real-time

5. Entity Relationship Diagram (ERD)

The database schema is built on PostgreSQL 16 with seven core entities. The users table serves as the central hub — all other tables maintain foreign key references back to users for ownership and audit tracking. The pgvector extension on ai_agent_logs enable embedding storage for Retrieval-Augmented Generation (RAG) in the Support Agent module.



5.1 Table: Users

Central user management table. All other entities reference this table via foreign keys.

Column Name	Data Type	Constraint / Notes	PK	FK	Index
id	SERIAL	PRIMARY KEY	**PK**		Yes
name	VARCHAR(100)	NOT NULL			
email	VARCHAR(150)	NOT NULL UNIQUE			Yes
password_hash	TEXT	NOT NULL			
role	ENUM(admin, manager, hr, support, viewer)	NOT NULL DEFAULT viewer			
department	VARCHAR(80)	NULLABLE			
is_active	BOOLEAN	DEFAULT TRUE			
last_login	TIMESTAMP	NULLABLE			
created_at	TIMESTAMP	DEFAULT NOW()			

5.2 Table: hr_Candidates

Stores candidate records uploaded by HR Officers. AI score and summary are populated by the HR Agent post-processing.

Column Name	Data Type	Constraint / Notes	PK	FK	Index
id	SERIAL	PRIMARY KEY	PK		Yes
uploaded_by	INTEGER	FK => users(id)		FK	
full_name	VARCHAR(120)	NOT NULL			
email	VARCHAR(150)	NULLABLE			Yes
phone	VARCHAR(30)	NULLABLE			
cv_file_path	TEXT	NOT NULL			
job_title	VARCHAR(120)	NULLABLE			
ai_score	DECIMAL(5,2)	CHECK 0-100			
ai_summary	TEXT	Gemini output			
skills	TEXT[]	Array of skill tags			
status	ENUM(pending,reviewed,shortlisted,rejected,hired)	DEFAULT pending			
created_at	TIMESTAMP	DEFAULT NOW()			

5.3 Table: Finance_invoice

Stores uploaded invoices. The Finance Agent populates anomaly_flag, is_duplicate, and ai_analysis fields.

Column Name	Data Type	Constraint / Notes	PK	FK	Index
id	SERIAL	PRIMARY KEY	PK		Yes
uploaded_by	INTEGER	FK => users(id)		FK	
vendor_name	VARCHAR(150)	NULLABLE			
invoice_number	VARCHAR(80)	UNIQUE			Yes
amount	DECIMAL(15,2)	NULLABLE			
currency	VARCHAR(10)	DEFAULT BDT			

Column Name	Data Type	Constraint / Notes	PK	FK	Index
invoice_date	DATE	NULLABLE			
due_date	DATE	NULLABLE			
status	ENUM(pending,processed, flagged,approved,rejected)	DEFAULT pending			
is_duplicate	BOOLEAN	DEFAULT FALSE			
anomaly_flag	BOOLEAN	DEFAULT FALSE			
anomaly_reason	TEXT	AI explanation			
ai_analysis	TEXT	Gemini output			
file_path	TEXT	Stored file path			
created_at	TIMESTAMP	DEFAULT NOW()			

5.4 Table: support_tickets
Customer support tickets. Sentiment and priority are determined by the Support Agent. ai_response is the RAG-generated reply.

Column Name	Data Type	Constraint / Notes	PK	FK	Index
id	SERIAL	PRIMARY KEY	**PK**		Yes
submitted_by	INTEGER	FK => users(id)		**FK**	
assigned_to	INTEGER	FK => users(id) NULLABLE		**FK**	
subject	VARCHAR(200)	NOT NULL			
body	TEXT	NOT NULL			
sentiment	ENUM(positive,neutral,negative,furious)	DEFAULT neutral			
priority	ENUM(low,medium,high,critical)	DEFAULT medium			
status	ENUM(open,in_progress,resolved,closed)	DEFAULT open			

Column Name	Data Type	Constraint / Notes	PK	FK	Index
ai_response	TEXT	RAG-generated response			
resolved_at	TIMESTAMP	NULLABLE			
created_at	TIMESTAMP	DEFAULT NOW()			

5.5 Table: security_events

Logs all security scan events. The Security Agent populates ai_verdict and confidence. Quarantined files are isolated.

Column Name	Data Type	Constraint / Notes	PK	FK	Index
id	SERIAL	PRIMARY KEY	**PK**		Yes
triggered_by	INTEGER	FK => users(id) NULLABLE		**FK**	
event_type	ENUM(phishing,spam,malware,injection,anomaly)	NOT NULL			
severity	ENUM(low,medium,high,critical)	DEFAULT medium			
source_ip	VARCHAR(50)	NULLABLE			
payload_hash	VARCHAR(64)	SHA-256 of content			
raw_content	TEXT	Original flagged content			
ai_verdict	TEXT	Gemini threat analysis			
confidence	DECIMAL(5,2)	0.00 to 100.00			
is_quarantine	BOOLEAN	DEFAULT			

Column Name	Data Type	Constraint / Notes	PK	FK	Index
d		FALSE			
resolved_at	TIMESTAMP	NULLABLE			
created_at	TIMESTAMP	DEFAULT NOW()			

5.6 Table: ai_agents_log
Audit log for all AI agent invocations. Stores token usage, latency, and embedding vectors for analytics and compliance.

Column Name	Data Type	Constraint / Notes	PK	FK	Index
id	SERIAL	PRIMARY KEY	**PK**		Yes
agent_type	ENUM(security,hr,finance,support,executive)	NOT NULL			
triggered_by	INTEGER	FK => users(id)		**FK**	
input_summary	TEXT	Truncated prompt summary			
output_summary	TEXT	Truncated response summary			
embedding	VECTOR(1536)	pgvector — semantic search			
tokens_used	INTEGER	Gemini token count			
latency_ms	INTEGER	Processing time			
status	ENUM(success,failed,timeout)	NOT NULL			
created_at	TIMESTAMP	DEFAULT			

Column Name	Data Type	Constraint / Notes	PK	FK	Index
		NOW()			

5.7 Table: executive_reports

AI-generated executive summaries produced by the Executive Agent. Each report covers a defined time period and module scope.

Column Name	Data Type	Constraint / Notes	PK	FK	Index
id	SERIAL	PRIMARY KEY	**PK**		Yes
generated_by	INTEGER	FK => users(id)		**FK**	
report_type	ENUM(daily,weekly,monthly,custom)	NOT NULL			
module_scope	TEXT[]	Array of included modules			
content	TEXT	Full AI-generated report			
period_start	DATE	Report start date			
period_end	DATE	Report end date			
created_at	TIMESTAMP	DEFAULT NOW()			

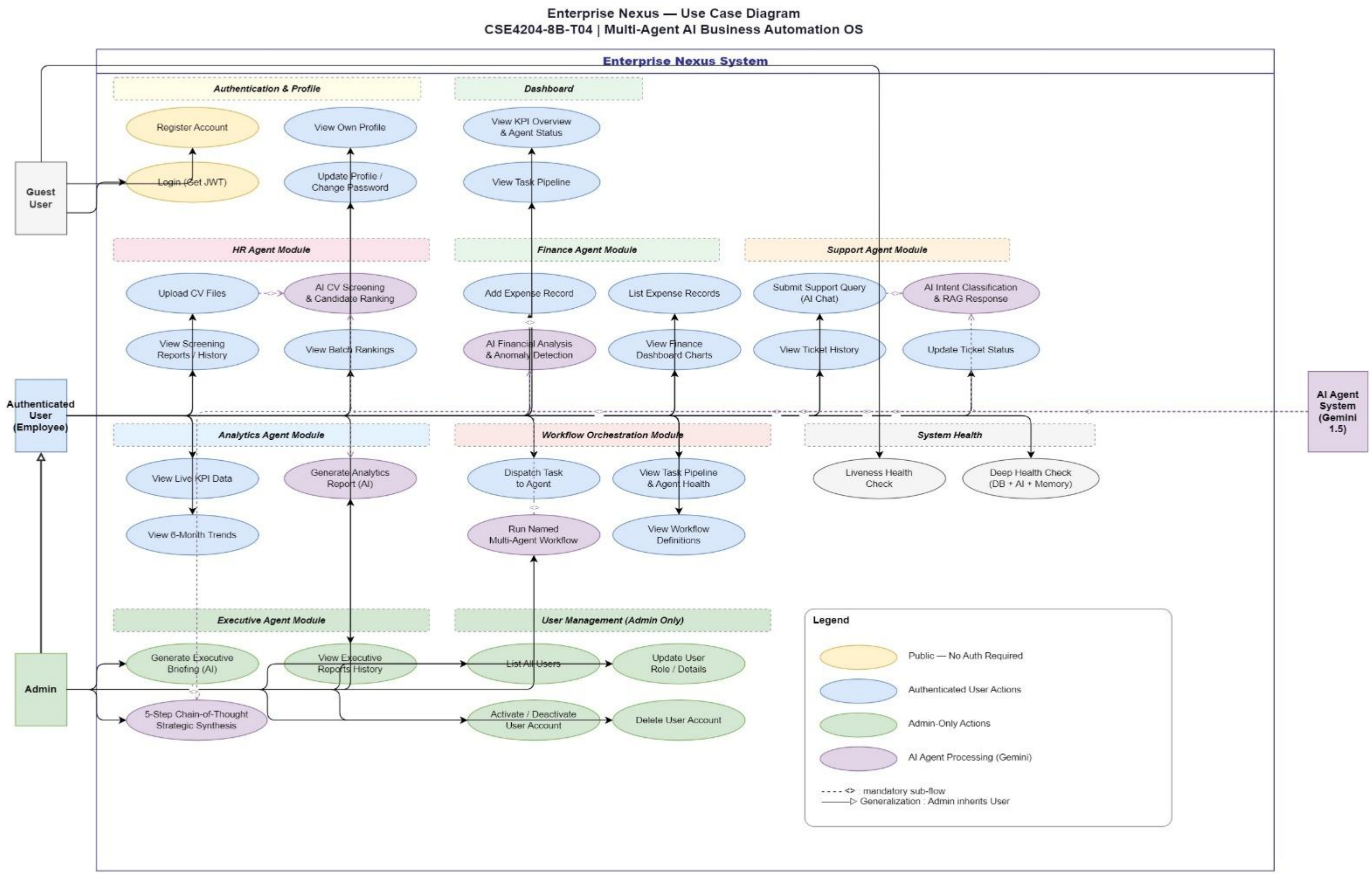
5.8 Relationship Summary

Relationship	Type	Cascade	Description
users -> hr_candidates	**1 : N**	SET NULL on delete	One user uploads many candidate CVs
users -> finance_invoices	**1 : N**	RESTRICT on delete	One user uploads many invoices
users -> support_tickets (submitted_by)	**1 : N**	RESTRICT	One user submits many support tickets
users -> support_tickets	**1 : N**	SET NULL	One user is assigned many tickets

Relationship	Type	Cascade	Description
(assigned_to)			
users -> security_events	1 : N	SET NULL on delete	One user's actions trigger many events
users -> ai_agent_logs	1 : N	RESTRICT on delete	One user's requests produce many logs
users -> executive_reports	1 : N	RESTRICT on delete	One user generates many reports

6. Use Case Diagram:

The use case diagram identifies four primary human actors and two system actors operating within the Enterprise Nexus system boundary. Use cases tagged with <<AI>> involve direct AI processing via Google Gemini API. The system boundary encompasses six functional modules.



6.1 Actors

Actor	Type	Access Level	Primary Use Cases
Admin	Human — Primary	Full System Access	User management, all module access, security oversight, report generation

Actor	Type	Access Level	Primary Use Cases
Manager	Human — Primary	**Analytics + Finance + Executive**	View dashboard, upload invoices, generate AI reports, NL query
HR Officer	Human — Primary	**HR Module**	Upload CVs, view candidates, schedule interviews, track applicants
Support Staff	Human — Primary	**Support Module**	Submit tickets, view ticket queue, read AI auto-responses
Google Gemini API	System — External	**AI Processing**	Auto-respond to tickets, generate reports, scan threats, analyze CVs
n8n Engine	System — Internal	**Orchestration**	Trigger agent workflows, route tasks, aggregate results, fire events

System Modules & Use Cases

A. Authentication Module — /api/v1/auth

Use Case	Actor(s)	AI?	Description
Register Account	All Actors	No	New users register with name, email, role
Login	All Actors	No	Authenticate and receive JWT token
Logout	All Actors	No	Invalidate current session token
View / Update Profile	All Actors	No	Read and update personal account details
Refresh Token	All Actors	No	Renew expired JWT token

B. HR Module — /api/v1/hr

Use Case	Actor(s)	AI?	Description
Upload CV	HR Officer, Admin	No	Submit candidate CV file for processing
AI Screen Candidate	HR Officer, Admin	**Yes (HR Agent)**	Trigger Gemini-powered CV analysis
View Candidate List	HR Officer, Admin	No	Browse filtered, sorted candidate records
Update Candidate Status	HR Officer, Admin	No	Move candidate through hiring pipeline

Use Case	Actor(s)	AI?	Description
Schedule Interview	HR Officer, Admin	No	Trigger automated interview scheduling email

C. Finance Module — /api/v1/finance

Use Case	Actor(s)	AI?	Description
Upload Invoice	Manager, Admin	**Yes (Finance Agent)**	Submit invoice; AI runs OCR and analysis
View Invoice List	Manager, Admin	No	Browse invoices with status and anomaly filters
View Anomaly Report	Manager, Admin	No	See flagged duplicate or suspicious invoices
Re-run AI Analysis	Manager, Admin	**Yes (Finance Agent)**	Force re-analysis on existing invoice
View Financial Summary	Manager, Admin	No	Aggregated spend stats by period

D. Support Module — /api/v1/support

Use Case	Actor(s)	AI?	Description
Submit Ticket	All Users	No	Create support request with subject and body
AI Auto-Respond	Support Staff, Admin	**Yes (Support Agent)**	Trigger RAG-based AI response generation
View Ticket Queue	All Users	No	Browse tickets with status and priority filters
Update Ticket Status	Support Staff, Admin	No	Assign, escalate, resolve, or close ticket
View Sentiment Analytics	Manager, Admin	No	Aggregated sentiment and emotion metrics

E. Security Module — /api/v1/security

Use Case	Actor(s)	AI?	Description
Scan Content	All Users (auto)	**Yes (Security**	Submit text or file for threat analysis

Use Case	Actor(s)	AI?	Description
		Agent)	
View Security Events	Admin	No	Browse all detected threats and verdicts
Quarantine File	Admin	No	Isolate flagged file in quarantine zone
Release Quarantine	Admin	No	Restore false-positive quarantined file
Live Alert Feed	Admin	No	Real-time WebSocket security alert stream

F. Executive Hub — /api/v1/executive

Use Case	Actor(s)	AI?	Description
View Dashboard	Manager, Admin	No	Real-time KPI cards, charts, agent status
Generate AI Report	Manager, Admin	**Yes (Executive Agent)**	AI synthesizes cross-domain executive summary
Natural Language Query	Manager, Admin	**Yes (Executive Agent)**	Ask business questions in plain English
View Report History	Manager, Admin	No	Browse previously generated reports
Cross-Module Analytics	Admin	No	Deep-dive analytics across all five modules

7. Activity Diagram:

Enterprise Nexus — AI Multi-Agent Workflow Activity Diagram

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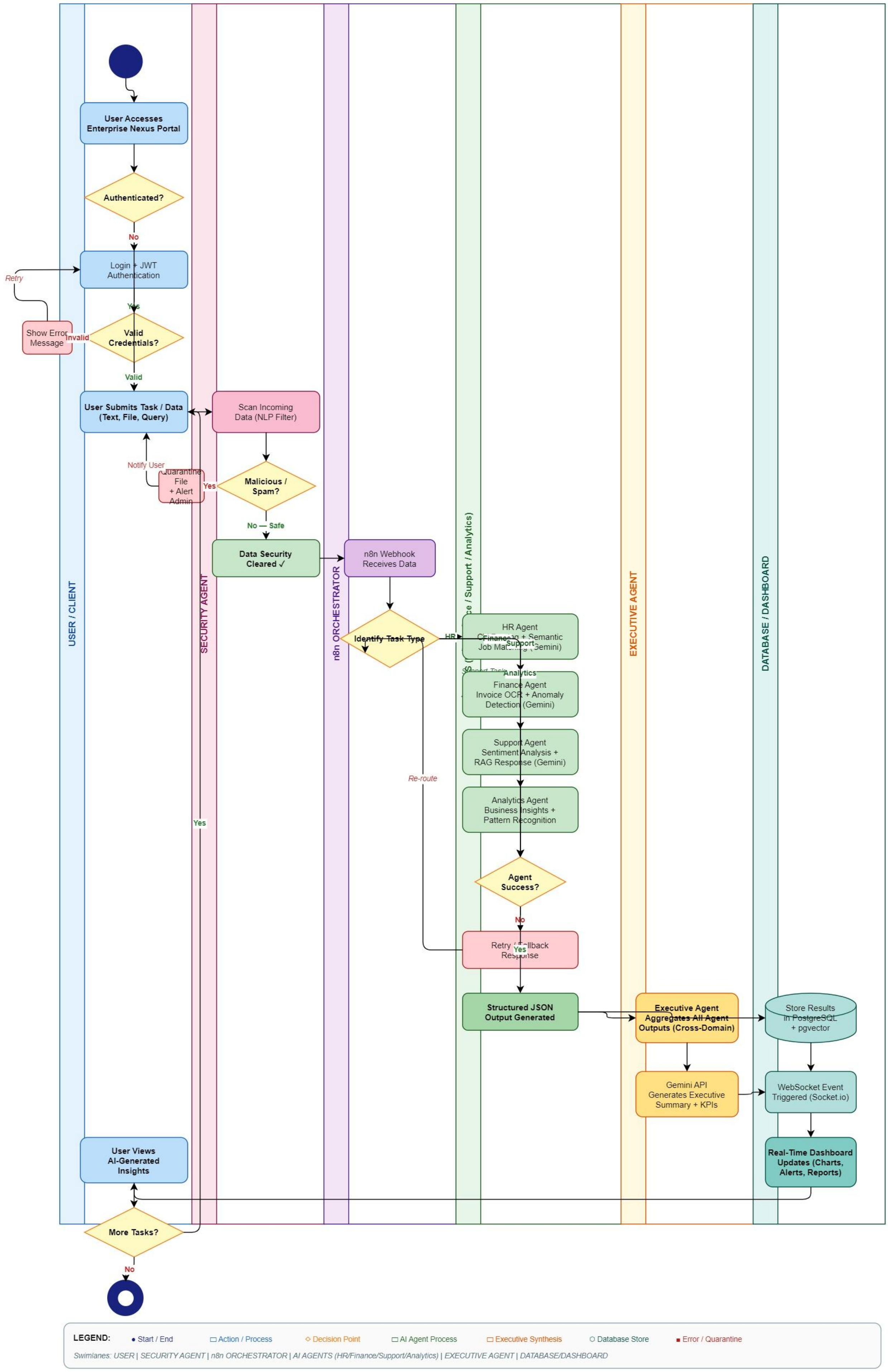


8. API Structure and End Point Reference

All endpoints follow RESTful design principles. Base URL: /api/v1. Every protected route requires a valid JWT Bearer token in the Authorization header. Responses follow a consistent JSON envelope: { success, data, message, timestamp }. WebSocket events are emitted on the /ws namespace for real-time dashboard updates.

Auth Level	Description
Public	No authentication required
Protected	Valid JWT required — any authenticated user
HR+Admin	Role must be hr or admin
Manager+	Role must be manager or admin
Admin	Role must be admin only
All Users	Any authenticated user regardless of role

8.1 AUTH MODULE — Base: /api/v1/auth

Method	Endpoint	Description	Auth	Request Body	Response
POST	/register	Register new user	Public	{ name, email, password, role }	{ token, user }
POST	/login	Authenticate and receive JWT	Public	{ email, password }	{ token, refreshToken, user }
POST	/logout	Invalidate current session	Protected	–	{ message }
GET	/me	Get current user profile	Protected	–	{ user }
PUT	/me	Update profile name and department	Protected	{ name, department }	{ user }
POST	/refresh	Refresh expired JWT	Protected	{ refreshToken }	{ token }
GET	/users	List all users	Admin	?role=&is_active=	{ users[] }
PUT	/users/:id/role	Update user role	Admin	{ role }	{ user }

8.2 HR MODULE — Base: /api/v1/hr

Method	Endpoint	Description	Auth	Request Body	Response
POST	/candidates	Upload CV and create candidate record	HR+Admin	multipart/form-data (cv_file, job_title)	{ candidate }
GET	/candidates	List candidates with filters	HR+Admin	?status=&sort=ai_score&order=desc	{ candidates[], count }
GET	/candidates/:id	Get full candidate detail	HR+Admin	–	{ candidate }
PUT	/candidates/:id	Update status or notes	HR+Admin	{ status, notes }	{ candidate }
DELETE	/candidates/:id	Delete candidate record	Admin	–	{ message }
POST	/candidates/:id/screen	Trigger AI CV screening	HR+Admin	–	{ ai_score, ai_summary, skills }
GET	/stats	HR pipeline statistics	HR+Admin	?period=	{ stats }

8.3 FINANCE MODULE — Base: /api/v1/finance

Method	Endpoint	Description	Auth	Request Body	Response
POST	/invoices	Upload invoice for AI processing	Manager+	multipart/form-data (invoice_file)	{ invoice }
GET	/invoices	List invoices	Manager+	?status=&anomaly=true&sort=	{ invoices[], count }
GET	/invoices/:id	Get	Manager	–	{ invoice }

Method	Endpoint	Description	Auth	Request Body	Response
		invoice with AI analysis	+		
POST	/invoices/:id/analyze	Re-run AI analysis	Manager +	–	{ analysis }
GET	/anomalies	Get all flagged anomalies	Manager +	?severity=	{ anomalies [] }
GET	/summary	Financial summary by period	Manager +	?period=monthly&year=2026	{ stats, chart_data }
GET	/duplicates	List detected duplicate invoices	Manager +	–	{ duplicates [] }

8.4 SUPPORT MODULE — Base: /api/v1/support

Method	Endpoint	Description	Auth	Request Body	Response
POST	/tickets	Create new support ticket	All Users	{ subject, body }	{ ticket }
GET	/tickets	List tickets with filters	All Users	?status=&priority=&sentiment=	{ tickets[], count }
GET	/tickets/:id	Get ticket with AI response	All Users	–	{ ticket }
PUT	/tickets/:id	Update status and assignment	Support+	{ status, assigned_to }	{ ticket }
POST	/tickets/:id/ai-respond	Trigger AI auto-response	Support+	–	{ response }
DELETE	/tickets/:id	Delete ticket (soft delete)	Admin	–	{ message }

Method	Endpoint	Description	Auth	Request Body	Response
GET	/sentiment	Sentiment analytics dashboard data	Manager +	?period=weekly	{ stats, chart_data }

8.5 SECURITY MODULE — Base: /api/v1/security

Method	Endpoint	Description	Auth	Request Body	Response
POST	/scan	Submit content for threat scan	All Users	{ content } or multipart (file)	{ verdict, confidence, event_id }
GET	/events	List security events	Admin	?severity=&type=&from=&to=	{ events[], count }
GET	/events/:id	Get full event detail	Admin	–	{ event }
PUT	/events/:id/quarantine	Quarantine flagged item	Admin	–	{ message }
PUT	/events/:id/release	Release quarantined item	Admin	–	{ message }
GET	/alerts	Live alert feed (WebSocket)	Admin	–	event: security_alert
GET	/stats	Security event statistics	Admin	?period=	{ stats }

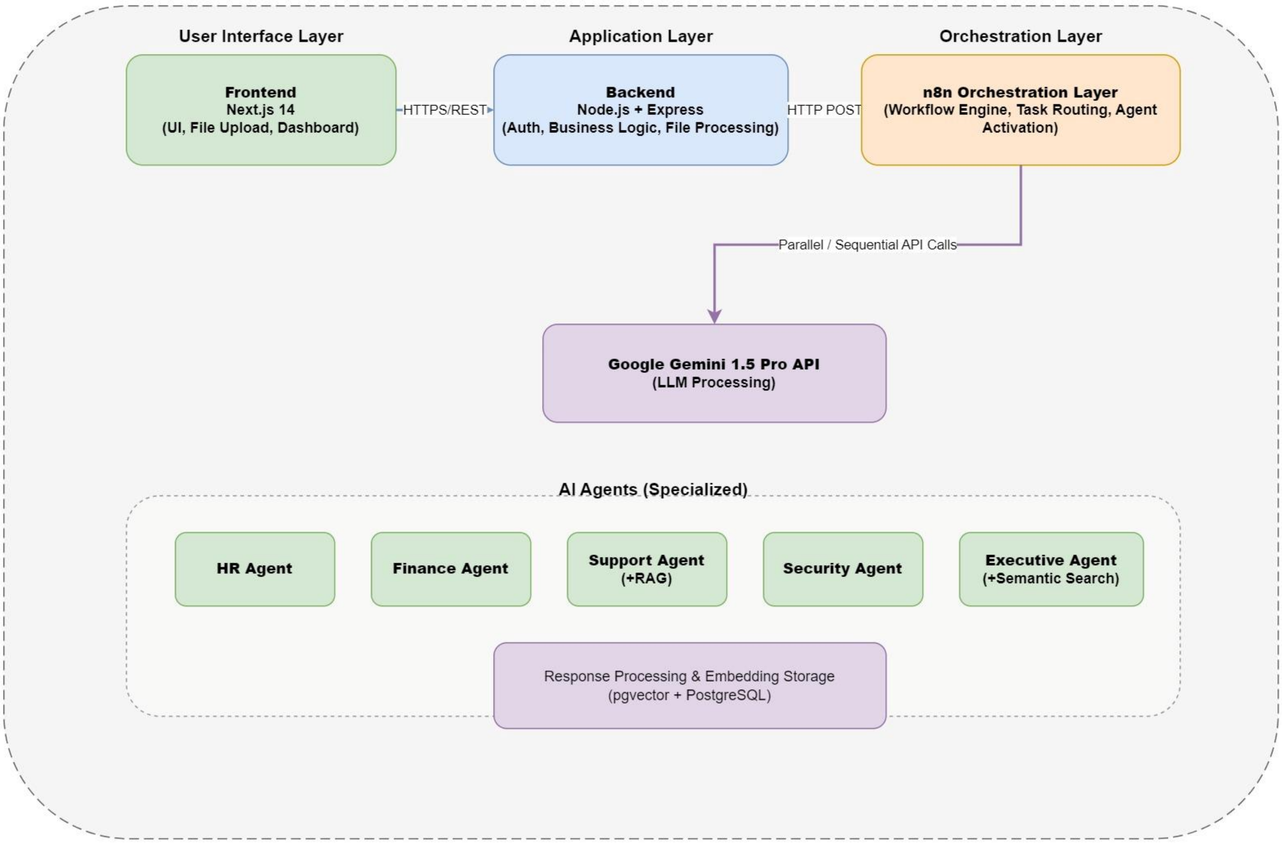
8.6 EXECUTIVE MODULE — Base: /api/v1/executive

Method	Endpoint	Description	Auth	Request Body	Response
GET	/dashboard	Real-time aggregated KPI data	Manager+	–	{ kpis, agent_status, alerts }
POST	/reports	Generate AI executive	Manager+	{ period_start, period_end, type, scope[] }	{ report }

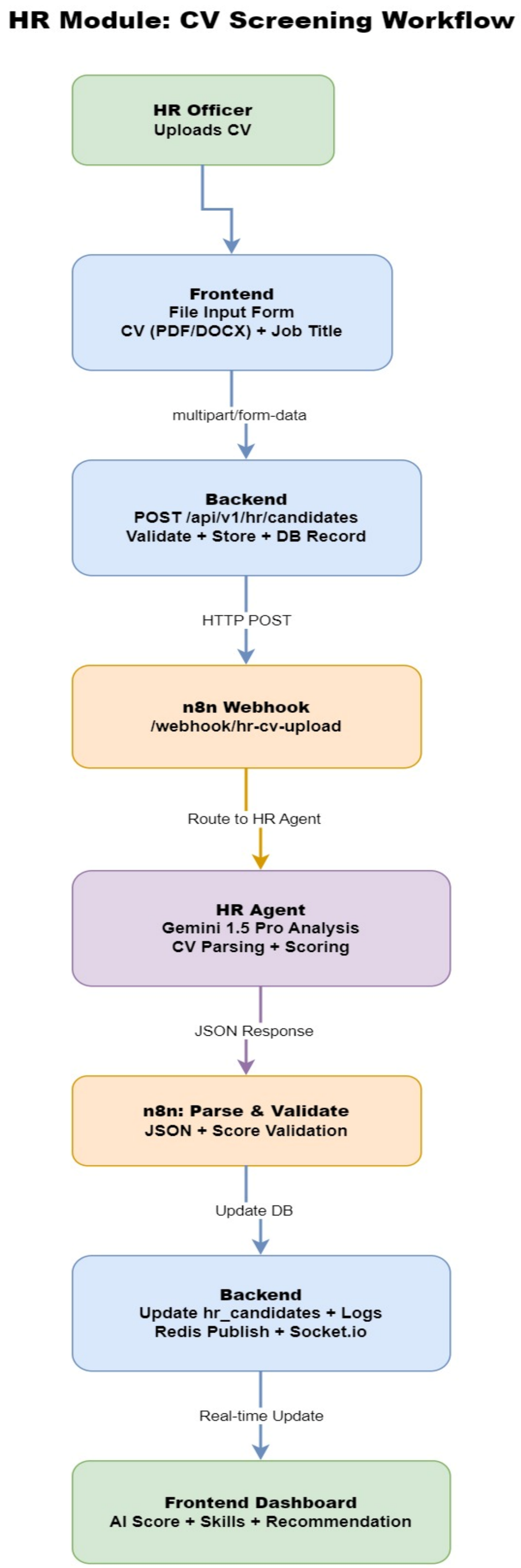
Method	Endpoint	Description	Auth	Request Body	Response
		report			
GET	/reports	List generated reports	Manager+	?type=&from=&to=	{ reports[] }
GET	/reports/:id	Get full report content	Manager+	–	{ report }
DELETE	/reports/:id	Delete report	Admin	–	{ message }
POST	/query	Natural language business query	Manager+	{ query_text }	{ answer, sources[], confidence }
GET	/analytics	Cross-module deep analytics	Admin	?module=&period=&metric=	{ analytics }

9. AI Workflow

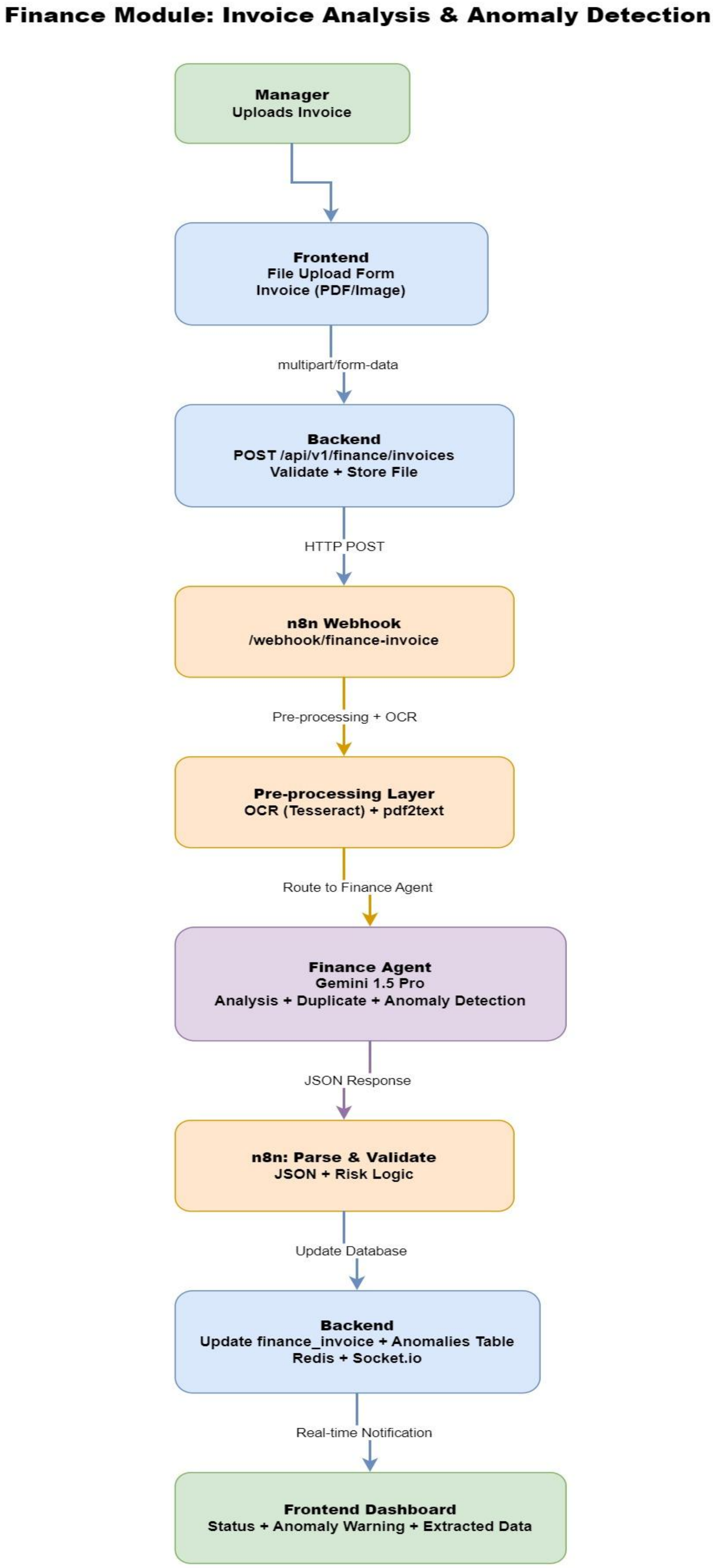
9.1 Agent Deployment Architecture



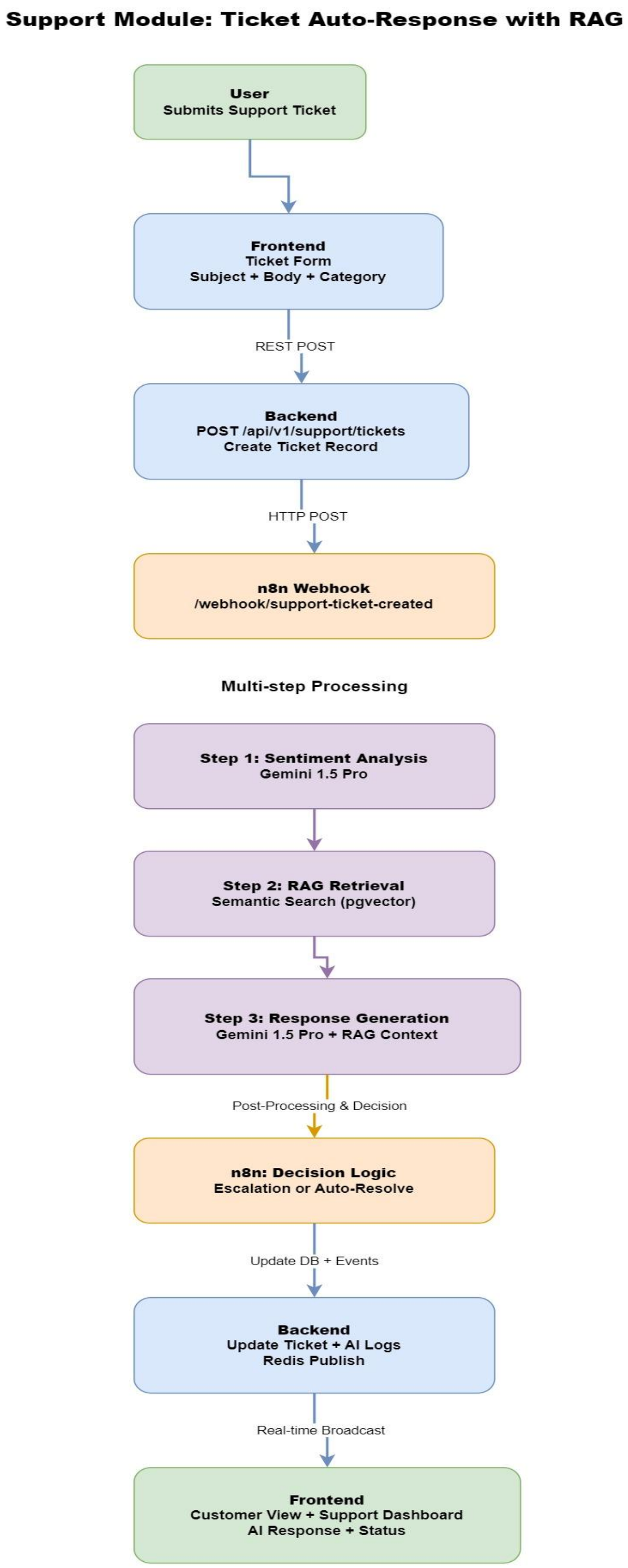
9.2 HR Module Flow Diagram



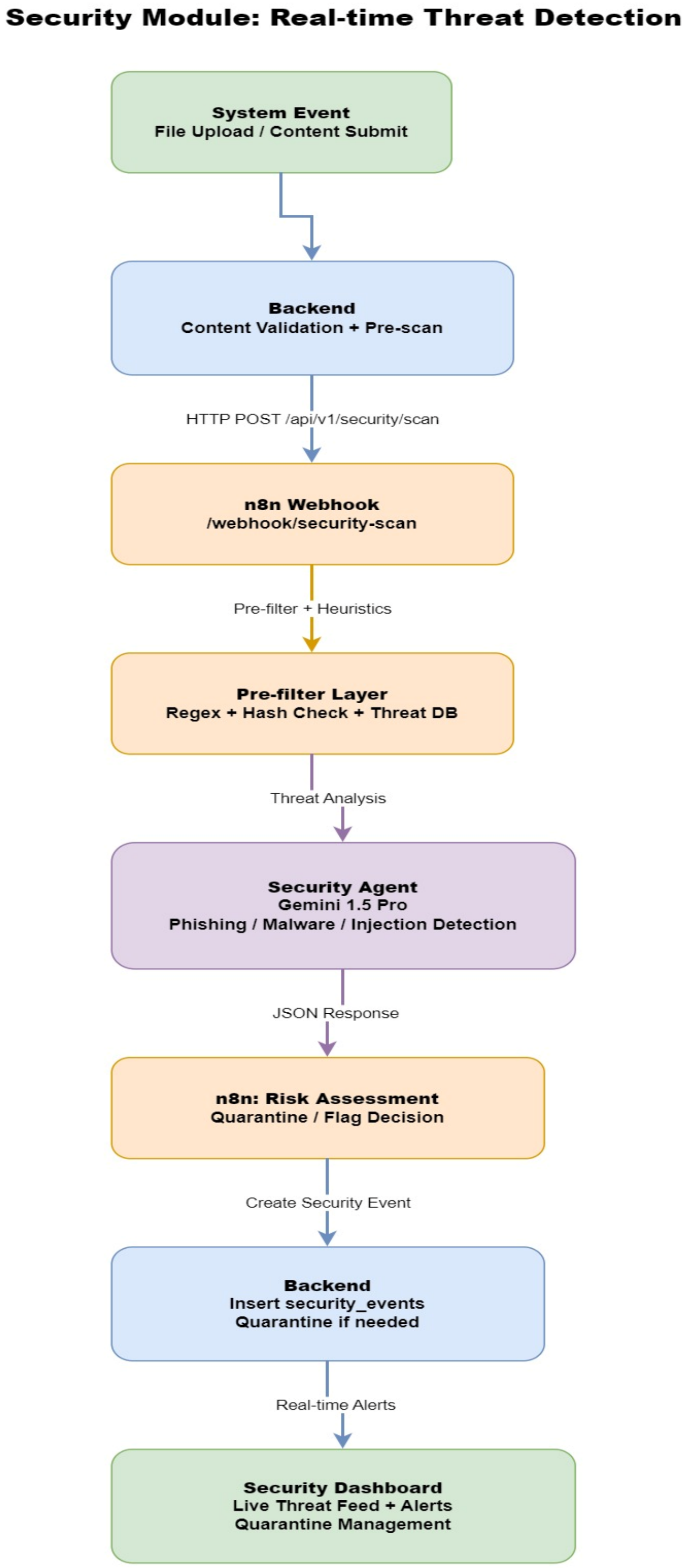
9.3 Finance Module Flow Diagram



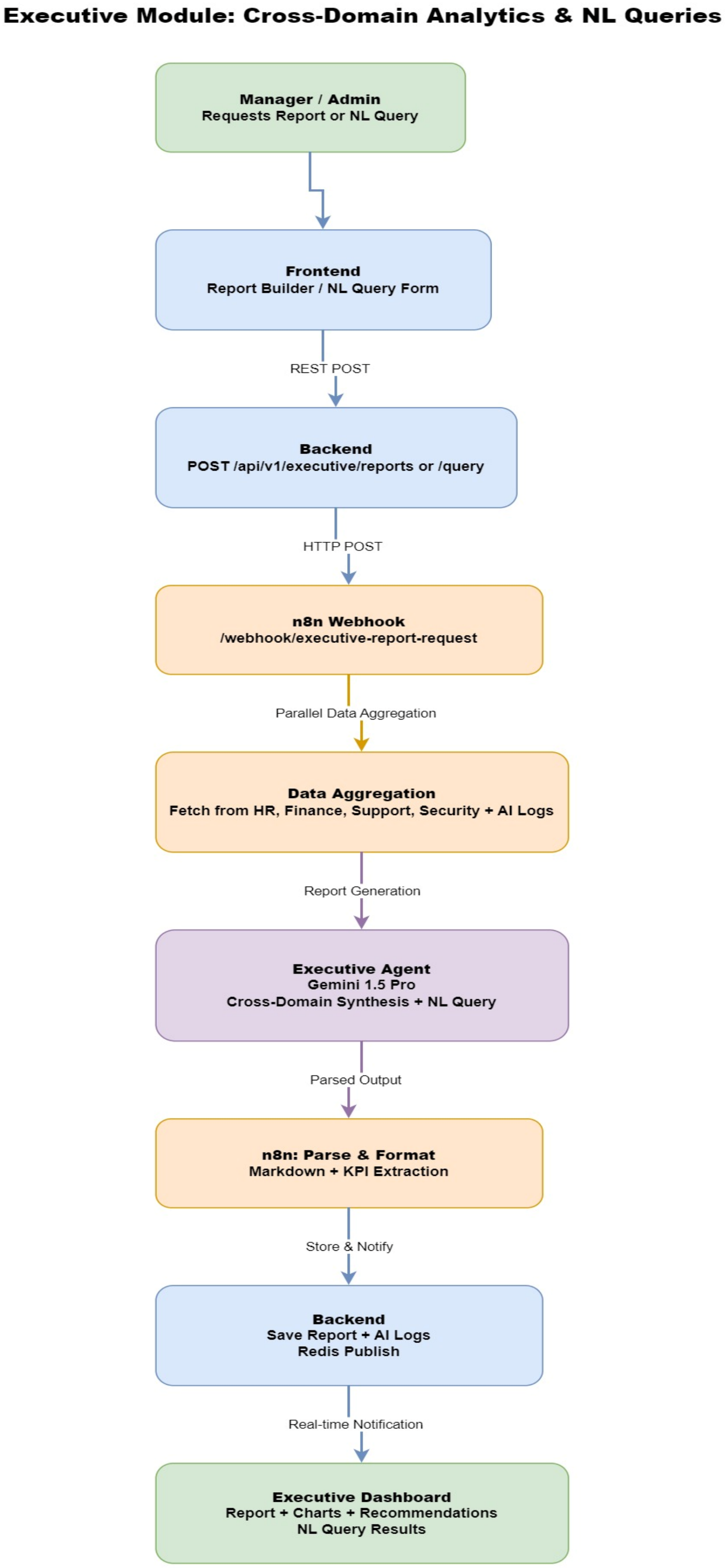
9.4 Support Module Flow Diagram



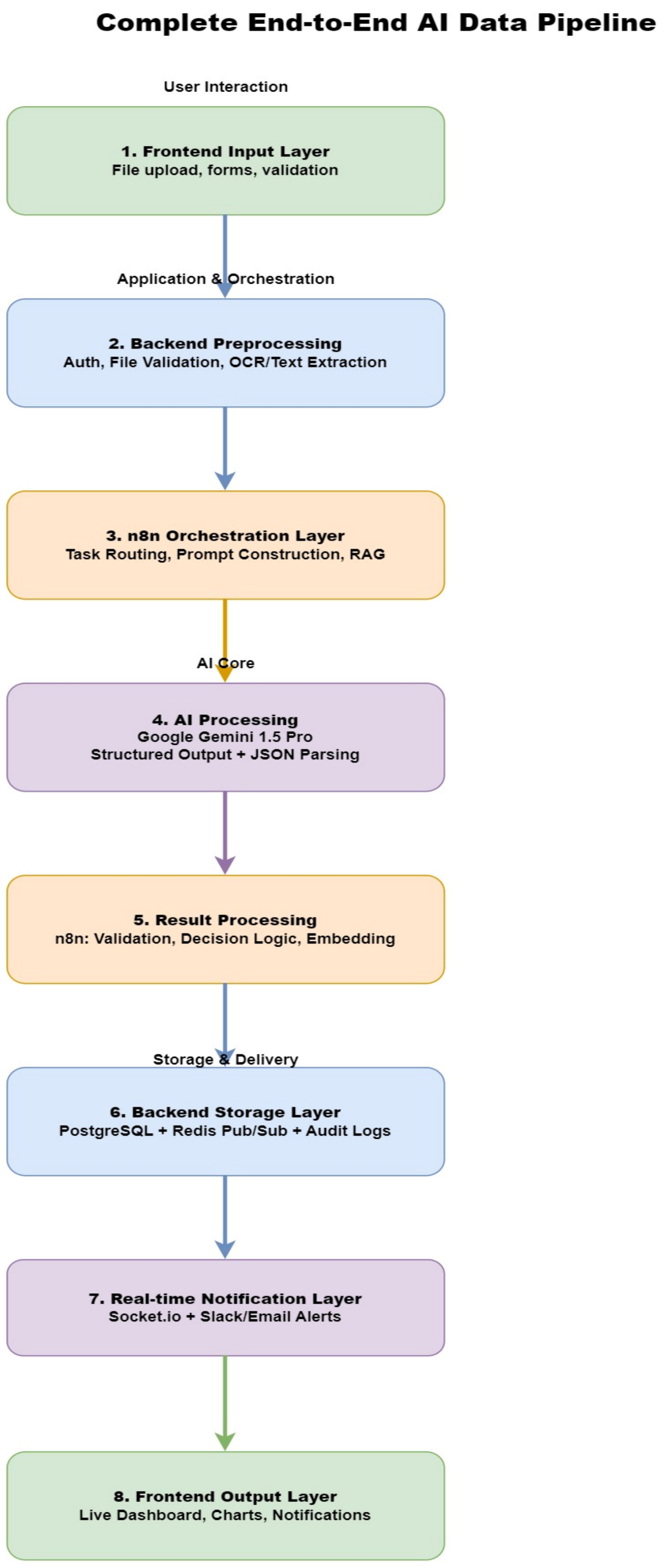
9.5 Security Module Flow Diagram



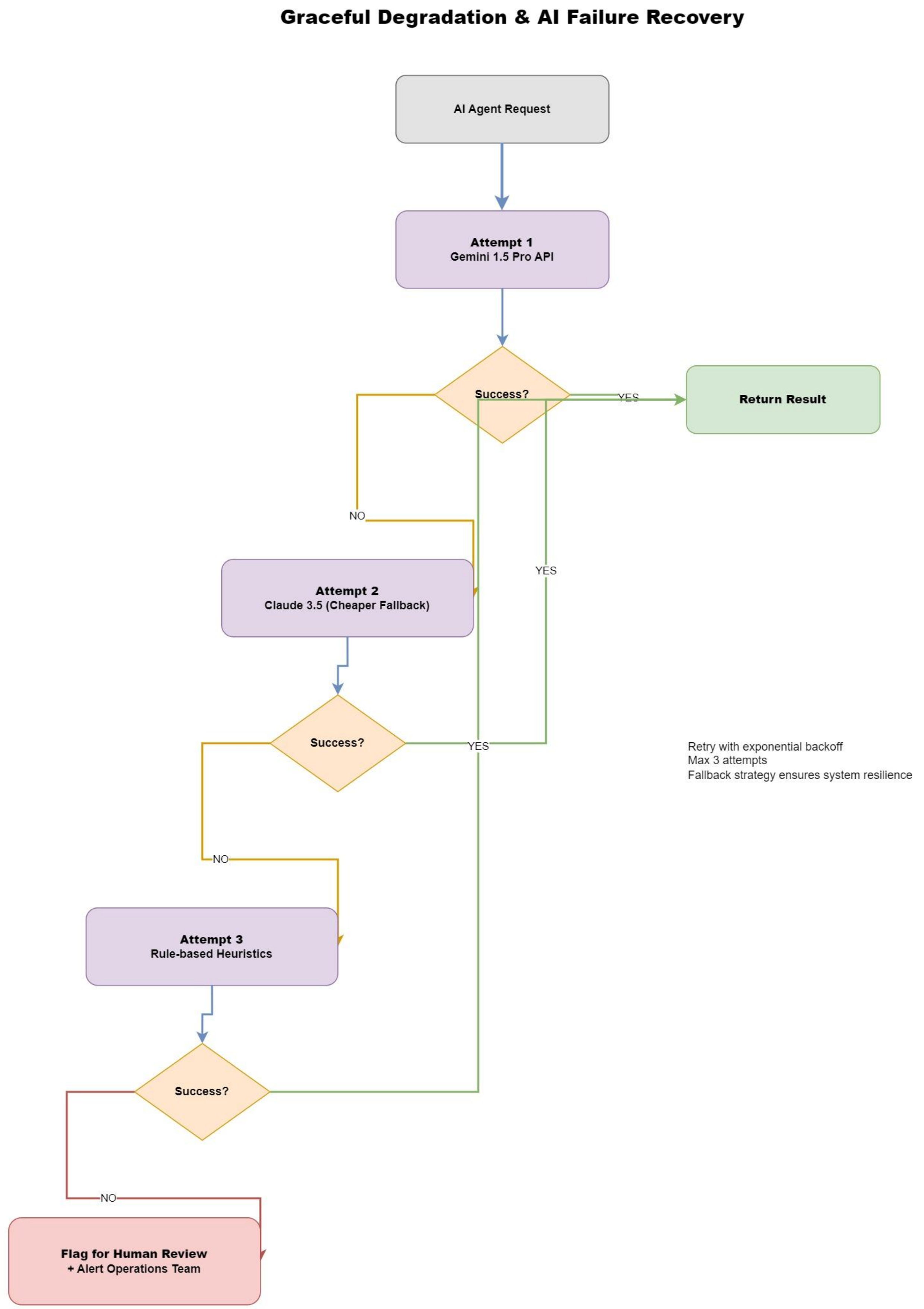
9.6 Executive Module Flow Diagram



9.7 Complete End-to-End Data Pipeline



9.8 Graceful Degradation Flowchart



10. References

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- Enterprise Nexus GitHub Repository: https://github.com/Nazmussakib247/CSE4204-8B-T04-Enterprise-Nexus